

Held-out value copying in small language models: a field-localized failure mode and the instrument to measure it

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Abstract

We study a specific faithfulness failure in small language models finetuned to convert short clinical dialogues into structured summaries: a **held-out copying gap** — the model copies field values it saw during finetuning but errs on held-out values under held-out phrasings, even though both are present verbatim in the input. In our own from-scratch models (3.15M and 10M parameters) this gap is large — 18.3 ± 1.3 and 18.7 ± 1.5 points of recall on the same multi-instance instrument used for the rest of the ladder (22–23 points on the single public instance the anchors were first scored on; §8). The failure is not diffuse: it localizes *entirely* to the three open-vocabulary fields (complaint, medication, allergy) and is **exactly zero** in the two closed-value fields (duration, severity), which act as a built-in control showing the effect is specifically held-out-*value* copying in open slots, not a generic degradation under unfamiliar phrasing. Finetuning the Pythia open-weight family (160M/410M/1B) on the identical task, the gap observed is **substantially smaller** (single-digit points; 3.5 ± 0.7 at 160M, 4.2 ± 0.9 at 410M, and an interval of $[0, 5]$ at 1B) — and that residual is itself field-localized: per-field re-scoring shows Pythia fully solves held-out complaints and medications, while its entire remaining gap is the one open slot with only five training values (allergy), where even 410M fails totally on the held-out value. We deliberately do **not** claim scale as the cause: the comparison also changes pretraining corpus, tokenizer, architecture, and finetuning method. The measurement itself is a second contribution. A pre-registered contamination check caught that our initial single-instance evaluation was under-powered for the gap; averaging over five larger evaluation instances resolved the middle rungs to sub-point precision. At 1B, a determinism cross-check then revealed that **training-run nondeterminism** — not evaluation noise — dominates the residual, so we report the top rung as an interval rather than a point. Both measurement lessons arose from empirical failures of the prior instrument, not from foresight.

1 Introduction

Production language systems for structured summarization (e.g. clinical scribing) are judged not on average-case benchmark scores but on faithfulness under distribution shift: whether the output contains only content grounded in the input, including on inputs unlike the training data (Maynez et al., 2020; Ji et al., 2023). We isolate one tractable component of this — value copying — in a setting where faithfulness is measurable by construction, and ask a single pre-registered question: **how does the copying gap behave as model capability increases?**

The contribution is threefold and we keep the parts distinct:

1. **Empirical.** A large held-out copying gap in sub-10M models is much smaller in the tested Pythia models, and localizes cleanly to the open-vocabulary fields (§8).

2. **Measurement.** Two lessons about estimating such a gap reliably: single-instance evaluation was under-powered (§6.1), and training nondeterminism bounds the estimate at the top of the ladder (§6.2).
3. **Engineering (companion).** A verification architecture that routes model errors to human review; its precision/review-load trade-off is measured in a companion write-up (in preparation), not in this paper.

2 Related work

Scaling laws and emergence. Language-model loss follows smooth power laws in parameters, data, and compute (Kaplan et al., 2020; Hoffmann et al., 2022), yet specific *task* behaviors can change sharply with scale: Wei et al. (2022) catalogue “emergent abilities” absent in small models and present in large ones. Schaeffer et al. (2023) argue much apparent emergence is a *measurement artifact* of discontinuous metrics (exact-match, multiple-choice) rather than a property of the model. Our study runs in the opposite direction — a behavior (the held-out copying gap) that is *large* in small models and *smaller* at larger scale — but the caution applies directly, since our gap is an exact-match recall difference; we therefore report it with across-instance error bars and, at 1B, separate metric behavior from training-run variance (§6.2). Where the emergence literature asks “does scale switch an ability *on*?”, we ask “does scale switch a faithfulness failure *off*?”, and find the honest answer is confounded with the training stack (§9).

The Pythia platform. Biderman et al. (2023) release Pythia, 16 models spanning 70M–12B trained on identical data in identical order, expressly to make scale a controlled variable. We use the 160M/410M/1B rungs as our scaling axis for exactly this property — within Pythia, architecture, tokenizer, pretraining corpus, and data order are fixed across sizes. The limitation we foreground (§9) is that our own-stack anchors are *not* in that controlled family, so the nano→Pythia step moves the stack as well as the scale — a confound Pythia’s internal control cannot remove.

Copying, pointer, and retrieval mechanisms. The behavior we measure is value *copying*: reproducing a field value present verbatim in the input. Explicit copy pathways have a long lineage in sequence-to-sequence models — CopyNet (Gu et al., 2016) and pointer-generator networks (See et al., 2017) add input-copying to abstractive generation. In transformers, induction heads implement a pattern-completion copy (Elhage et al., 2021; Olsson et al., 2022), and *retrieval heads* — a sparse (<5%), apparently universal set of attention heads — copy tokens from context into the output and mechanistically explain long-context factuality (Wu et al., 2024). These are the natural mechanistic hypotheses for our gap: a model that copies *seen* values but fails on *held-out* values under held-out phrasing may have copy/retrieval circuitry that generalizes poorly off-distribution. We deliberately defer this question (§10) until the behavioral phenomenon is deconfounded, so any circuit we implicate is attached to a clean empirical contrast rather than a stack difference.

Faithfulness and hallucination. Faithfulness — output grounded only in the input — is a central failure mode of abstractive summarization; Maynez et al. (2020) distinguish intrinsic from extrinsic hallucination, and Ji et al. (2023) survey the phenomenon across NLG. Most such work relies on human judgment or model-based entailment scoring. Our task is constructed so faithfulness is *exact*: because each dialogue is rendered from a known fact tuple, a summary is scored field-by-field against ground truth with no judge model, and our “hallucination” is precisely an extrinsic

fabrication of a field value. This trades ecological realism for a noise-free, reproducible faithfulness signal.

Reproducibility and training nondeterminism. Run-to-run variation from seeds and low-level tooling can be large enough to reverse method rankings (Picard, 2021; Pham et al., 2020), and even fixed-seed training is nondeterministic through non-associative GPU reductions and library tooling (Zhuang et al., 2022); the standard prescription is to report mean $\pm$ SD over multiple runs. Our 1B result is a concrete downstream instance: two fixed-seed retrains of the same model produced gaps of 5 and 0 on the byte-identical evaluation, so at that rung the *unit of uncertainty* shifts from the evaluation instance to the training run, and we report an interval (§6.2). This connects the reproducibility literature to a specific measured behavior (a faithfulness gap) rather than to top-line accuracy.

Evaluation reliability and contamination. Public benchmarks risk contamination — memorized test content inflating scores (Xu et al., 2024) — with held-out or rolling test data the standard defense. Our benchmark is a *seeded generator* rather than a fixed set, so fresh evaluation instances can be drawn at will, and our pre-registered contamination check compares the public instance against fresh draws. We find the public instance is *harder* (higher gap) than fresh draws at every rung — the opposite of the contamination signature — which both rules out memorization and surfaces a distinct hazard: a fixed public instance can be a systematically biased difficulty draw even when content-addressed and reproducible.

3 Task and benchmark

Faithfulness is exact by construction. Synthetic clinic dialogues are rendered from a known fact tuple of five fields (chief complaint, duration, severity, medication, allergy), so each summary is scored field-by-field against ground truth with no judge model — a noise-free faithfulness signal, traded against ecological realism.

The held-out copying gap. Evaluation dialogues are held-out along two axes — held-out *template families* (all dialogues) and held-out *slot values* (half the dialogues) — and the central object is the **held-out copying gap**: seen-value recall minus held-out-value recall under held-out templates, which isolates value copying from phrasing familiarity.

The benchmark is a generator. The scientific object is the eval *distribution* (a seeded generator), not any fixed set of dialogues. Fresh instances can be drawn to defend against memorization of any specific instance (§6.1), and the instance-difficulty distribution can be measured directly — a property we exploit in both measurement lessons.

4 Prior stages (setup)

Two earlier results frame the question. **Stage C** tested whether the gap is a training-curriculum artifact by adding an unmemorizable-value training slice; the held-out gap was unmoved, refuting the curriculum explanation. **Stage S** scaled our own stack from 3.15M to 10M parameters: the larger model passed the average-case faithfulness gate (parse 100%, recall 88%, hallucination 7.5%) yet its held-out gap was essentially unchanged (≈ 23 vs. ≈ 22 points single-instance; 18.7 ± 1.5

vs. 18.3 ± 1.3 on the multi-instance instrument, §8 — the near-identity now measured with error bars, not two single points). Stage S concluded that a $3.2\times$ scale step does not cross the failure — motivating a wider ladder.

5 Stage T design (pre-registered)

Ladder. Own models at 3.15M/10M (re-scored on the multi-instance instrument, §8) plus the Pythia family at 160M/410M/1B, finetuned on the identical scribe recipe with LoRA and scored by the identical faithfulness scorer (full configuration in §7).

Why Pythia. The question is scaling behavior, not model ranking. Pythia (Biderman et al., 2023) holds architecture, tokenizer, pretraining corpus, and data order fixed across sizes, so within the family relatively few variables move with parameter count — the property a scaling study needs. This choice does not isolate scale from the nano→Pythia *stack* change (§9); it is the reference family for that axis, not a claim that scale is the cause.

Pre-registration and falsifiers. Bars, bands (PERSISTS/THRESHOLD/DIVERGENT), seeds, and decision rules were fixed before measurement (three pre-measurement amendments). A base-model control (un-finetuned model must fail parse $< 50\%$) guards against a trivially-solvable task; it passed at every rung (base parse 0%). The measured code was adversarially reviewed before any run; three measurement-invalidating defects were fixed pre-measurement.

6 The measurement instrument, and why it changed

The instrument evolved because the data demanded it. We present this as a result.

6.1 Single-instance evaluation was under-powered

The pre-registered contamination check (score the same model on the public instance vs. a fresh instance; require per-metric agreement within 5 points) **flapped** at the threshold across rungs (gap differences $\approx 5, 6, 4$). We attribute this to the gap riding on a handful of hard held tokens, so a single 20-dialogue instance carries ± 5 –6 points of sampling variance. Crucially, the *direction* of the discrepancy (the public instance was **harder**, not easier) is the opposite of the contamination signature the check was built to detect — ruling out memorization. The fix re-scores the same frozen models on **five instances of 100 held dialogues each** and reports the gap as a mean $\pm$ across-instance SD. This cut the SD to ~ 0.7 – 0.9 points at 160M/410M. We applied the identical re-scoring to the own-stack anchors so the whole ladder shares one instrument (re-scoring only — frozen checkpoints, their native ChatML/greedy scorer; both anchors reproduced their canonical single-instance readings byte-for-byte before the multi-instance pass). The anchors’ single public-instance gaps (22.4, 23.0) were themselves modestly hard draws — ~ 4 points above their multi-instance means (18.3, 18.7) — the **same direction** (public instance harder, not easier) found at the Pythia rungs, so the public instance is a systematically hard draw across the entire ladder.

6.2 Training nondeterminism bounds the 1B estimate

The re-scoring re-generates each frozen adapter by re-finetuning at the fixed seed; we verify this reproduces the original model by re-scoring the byte-identical original instances and checking the

gaps match (a **determinism cross-check**). At 160M and 410M the match was exact. At 1B it **failed**: on the byte-identical instance, one training run gave a 5-point gap and another gave 0. With evaluation held constant, the difference is training-run variance — fp16 with non-associative GPU reductions over ~ 1125 steps of a 1B model that, we conjecture, sits near the boundary of perfect held-out copying — the tooling-level nondeterminism documented by Zhuang et al. (2022) (cf. Picard, 2021, on seed variance exceeding effect sizes), here surfacing in a downstream faithfulness metric rather than top-line accuracy. The right instrument for the 1B gap is therefore multiple training seeds, not multiple evaluation instances; we report an interval. With only two retrains we bound the *magnitude* of training-run variance at this rung (it is large — comparable to the whole residual gap) but cannot characterize its distribution; the interval spans the two observed outcomes rather than a modelled spread, and a fuller characterization (how often a run tips to perfect held-out copying) would need more seeds.

7 Methods (consolidated protocol)

Datasets (a seeded generator). Each example is a synthetic doctor–patient dialogue rendered from a fact tuple of five fields — chief complaint (cc), duration (dur), severity (sev), medication (med), allergy (alg) — paired with the target summary **CC**: ... | **DUR**: ... | **SEV**: ... | **MED**: ... | **ALG**: Training values (seen): a fixed complaint pool plus a compositional body-part $\times$ sensation pool (~ 190 complaints, so cc cannot be solved as closed-set classification), 18 medications, 5 allergies, 3 severities. Finetuning data: 12,000 examples, full supervised finetune, 3 epochs, LR 10^{-4} (own stack); the eval distribution is a separate generator.

Held-out protocol (two axes). Evaluation dialogues are held-out along two independent axes. (i) *Held template families*: doctor/patient surface phrasings never seen in finetuning — used by **all** eval dialogues. (ii) *Held slot values*: six specific values excluded from all finetuning data (complaints {toothache, neck pain, heartburn}, medications {melatonin, throat lozenges}, allergy {sulfa drugs}) — used by **half** the eval dialogues; the other half draw seen values. The **held-out copying gap** = seen-value recall – held-out-value recall, both measured under held templates, isolating value copying from phrasing familiarity.

Model families and finetuning. *Own stack*: from-scratch GPT (RoPE, GQA, SwiGLU, RM-SNorm pre-norm, tied embeddings, 4098-vocab BPE, 512 ctx) at 3.15M ($d=192, L=6, H=6, KV=2, ff=512$) and 10M ($d=320, L=8, H=8, KV=2, ff=864$), pretrained on ~ 200 M FineWeb tokens ($D \approx 20N$; Hoffmann et al., 2022) and full-finetuned on the scribe task; the two anchor checkpoints are frozen release assets. *Pythia*: EleutherAI pythia-{160m,410m,1b}, adapted with LoRA ($r=16, \alpha=32$, dropout 0, targets {query_key_value, dense, dense_h_to_4h, dense_4h_to_h}), LR 10^{-4} , 3 epochs.

Evaluation metric. Greedy decoding; the output is parsed against the fixed field template. A field is a *hit* (exact match), an *omission* (“none” for a present value), or a *hallucination* (otherwise). Held/seen recalls are computed over the fields of **parsed** dialogues only (identical convention across stacks). Two decode paths, each matched to its stack’s native format: own stack uses ChatML with a token-by-token argmax loop stopping on the end token, max 64 new tokens; Pythia uses raw text with greedy **generate**, EOS stop, max 64 — these are the checkpoints’ own inference formats, not a shared decoder.

Model	Params	Stack	Gap (5×100 held)	inst0
nano	3.15M	own	18.3 ± 1.3	22.4
scale	10M	own	18.7 ± 1.5	23.0
pythia-160m	160M	Pythia	3.5 ± 0.7	7.0
pythia-410m	410M	Pythia	4.2 ± 0.9	8.0
pythia-1b	1B	Pythia	$[0, 5]^\dagger$	5.0/0.0

Table 1: Held-out copying gap across the ladder on one multi-instance instrument (mean $\pm$ across-instance SD over five 100-held instances). inst0 is the single public instance. † 1B is bounded by training-run nondeterminism, not eval noise.

Multi-instance evaluation. The powered instrument is $K=5$ fresh evaluation instances (fixed seeds), each 100 held + 100 seen dialogues. A model’s gap is the mean over the five instances; its uncertainty is the across-instance SD. The single public instance (40 dialogues) and an auxiliary instance are retained for the cross-checks below.

Determinism checks (why “re-score”, not “re-run”). All measurements are re-scorings of frozen models. Before the multi-instance pass, each model is re-scored on the canonical single instance and required to reproduce its archived reference: the own-stack anchors reproduced their reference logs exactly (one even across a CUDA→MPS device change). For Pythia, re-finetuning at the fixed seed regenerates the frozen adapter (headless-GPU reproduced interactive-GPU byte-for-byte), verified in-band by requiring the re-scored instance gaps to match the archived originals. A check that fails (as at 1B) is treated as a finding, not smoothed over.

Statistical reporting. Powered rungs report gap mean $\pm$ across-instance SD (ddof=1); each Pythia instance additionally carries a 95% bootstrap CI (10,000 resamples over dialogues). The contamination check is direction-aware — it flags only if the public instance gap is *below* the fresh-instance mean by more than 2 SD (memorization would make the public instance easier); a higher public gap is a hard draw, not contamination. The 1B rung is reported as an interval $[0, 5]$ from two fixed-seed training-run retrains, not as a point.

8 Results

8.1 The gap across the ladder

All five rungs are scored on the **same multi-instance instrument** (five instances of 100 held + 100 seen dialogues, gap = mean $\pm$ across-instance SD). The own-stack anchors’ original single-instance readings are shown alongside for provenance.

Model-side quality is high at every rung (own-stack anchors parse 94–100%, recall 80–91%, hallucination 7–12%; Pythia rungs parse 100%, recall 96–100%, hallucination 0–4%), on both the public and fresh instances. On this single consistent instrument the large held-out copying gap **observed in our own-stack models (~ 18 points) was substantially smaller in the tested Pythia models (3.5–4.2 points)**. The reduction (~ 14 – 15 points) is an order of magnitude larger than any noise source identified (anchor SD ≈ 1.3 – 1.5 , Pythia SD ≈ 0.7 – 0.9), so the direction is not in question. Note that the single-instance column — the basis of the earlier headline — is a uniformly hard draw (higher gap at every rung), which is precisely why mixing a single-instance

Field	held-out values?	nano gap	scale gap
cc (chief complaint)	yes (3 held)	45.3 ± 8.1	58.2 ± 3.5
med (medication)	yes (2 held)	23.8 ± 4.8	14.2 ± 2.3
alg (allergy)	yes (1 held)	22.5 ± 4.5	21.2 ± 4.5
dur (duration)	no (numeric)	0.0	0.0
sev (severity)	no (closed 3-set)	0.0	0.0

Table 2: Dialogue-level (“diluted”) per-field gap, mean $\pm$ SD over the five instances.

anchor with a multi-instance Pythia number would have overstated the contrast; on the consistent instrument the gap is still large but the anchors read ~ 18 , not ~ 22 – 23 .

Where the gap lives (fieldwise). Breaking the own-stack anchor gap down by field (re-scored on the same five instances) localizes it sharply (Table 2).

The entire gap sits in the three fields that *have* held-out vocabulary values (cc, med, alg) and is **exactly zero** in the two whose values are all in-distribution (dur is numeric; sev is the closed set {mild, moderate, severe}). The two zero-gap fields are a **template-vs-value control**: dur and sev undergo the *same* held-out template shift as every other field (all eval dialogues use held-out phrasings) yet show no gap, so the gap is driven by held-out *values*, not unfamiliar phrasing. (This is not a claim that the model is robust to held-out dur/sev *values* — there are none; the closed fields isolate the template axis from the value axis.) Two magnitude caveats follow from the fact that the seen/held split is at the *dialogue* level: (i) within a “held” dialogue only $\sim 68\%$ of cc fields actually carry a held-out value, and just $\sim 30\%$ of med and $\sim 21\%$ of alg fields do (the rest are seen values or “none”; fractions are $P(\text{field carries a held-out value} \mid \text{held dialogue})$, measured over the five instances), so the held buckets for cc/med/alg are *diluted* with seen values — the reported per-field gaps are therefore conservative and **not directly comparable across fields** (the true held-out-value gap is larger, and most diluted for med/alg); and (ii) for the same reason the aggregate ~ 18 is a lower bound on the pure held-out-value effect. Neither caveat affects the cross-model comparison, since every rung is scored on the identical instrument. The split still explains the anchors’ aggregate near-identity (18.3 vs. 18.7) masking field-level differences — nano and scale trade off cc against med — and identifies chief complaint, with its ~ 190 compositional values, as the hardest copy.

The Pythia residual is also field-localized — to a single field. Re-scoring the Pythia rungs per-field on the same instrument (adapters regenerated deterministically; the built-in comparability check reproduced the published per-instance gaps **exactly**, max diff 0.0) shows the diluted per-field gap is ≈ 0 for cc, dur, sev, *and med* at both rungs — the **entire** residual sits in allergy (160M: 17.6 ± 3.3 ; 410M: 21.2 ± 4.5 ; divided over the five fields these reproduce the 3.5/4.2 aggregates). On the clean value-level metric the contrast is stark: cc **0.0** and med **0.0** at both rungs — held-out-value copying for complaints and medications is *solved* — while the single held-out allergy value fails **$83.6 \pm 4.6\%$** of the time at 160M and **$100.0 \pm 0.0\%$** at 410M, the same total failure as the anchors, and it does not improve with scale within Pythia (it worsens slightly). The clean ladder is therefore: anchors 87.3/79.5 $\rightarrow$ Pythia **14.7 ± 2.1** (160M) / **17.7 ± 3.2** (410M), with the residual now *interpretable* rather than diffuse.

At 1B the comparability self-check **failed exactly as §6.2 predicts**: the regenerated adapter is a *different training draw* (per-instance aggregates 0.6–2.2 vs. the published draw’s all-zeros — inside

Field	nano 3.15M	scale 10M	pythia-160m	pythia-410m
cc	71.8 ± 4.9	87.4 ± 3.5	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
med	100.0 $\pm$ 0.0	47.1 ± 4.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
alg	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	83.6 $\pm$ 4.6	100.0 $\pm$ 0.0
aggregate (cc+med+alg)	87.3 $\pm$ 2.7	79.5 $\pm$ 2.1	14.7 $\pm$ 2.1	17.7 $\pm$ 3.2

Table 3: Value-level (“clean”) per-field gap among value-bearing items, mean $\pm$ SD over the five instances — the clean ladder. Pythia solves cc and med outright; the entire residual is the five-training-value allergy slot.

the $[0, 5]$ interval), so its fieldwise numbers describe a third training draw, not the published 1B rung, and we do not place them in the ladder. The pattern, however, is unchanged: clean cc and med are 0.0 in this draw too, with the entire residual again in allergy (clean 24.6 ± 10.3 ; diluted 5.4 ± 3.2). The single-slot localization is thus stable across training draws even in the nondeterministic regime — and the self-check’s loud failure is the §6.2 lesson operating as designed.

A slot-diversity hypothesis (labeled as such). The three open slots differ sharply in training-value diversity — ~ 190 complaints, 18 medications, 5 allergies — and the failure tracks it inversely at every scale tested: the 3M anchor fails all three; the 10M anchor partially recovers medications; Pythia solves complaints and medications yet still fails the allergy slot totally. A slot with only five training values is learnable as a closed-set classifier, so no copy mechanism need ever be induced for it — whereas ~ 190 compositional complaints force copying. We advance this as a hypothesis (*per-slot training diversity, not model capability alone, governs whether copy or classify wins*), directly testable by varying slot diversity at fixed scale; it also predicts the allergy-slot failure would persist in far larger models trained on this recipe.

The undiluted failure (clean metric). Removing the dilution — scoring each field’s *value-bearing* items by whether the value is actually held-out vs. seen, with the template held fixed — gives the *pure* held-out-value copying gap, and it is severe (Table 3).

On *actual* held-out **allergy** values (both anchors) and held-out **medications** at 3M, the models recall $\approx 0\%$ while copying seen values $\sim$ perfectly — a ~ 100 -point gap; the 10M model is the one place scale visibly helps, recovering about half of its held-out medications (a 47-point gap). Aggregated over the value fields the pure held-out-value gap is ~ 80 –**87 points**, not the diluted 18 — a near-total copy failure the dialogue-level metric substantially understates. To keep this honest we **distinguish two metrics** and do not mix them: the *dialogue-level* (“diluted”) gap is the ladder spine — it is available for every rung on the identical instrument — while the *value-level* (“clean”) gap is reported wherever the per-field re-score has run: the anchors here and the Pythia 160M/410M rungs below (the 1B re-score landed a different training draw — see below); clean is only ever compared with clean, diluted with diluted.

What the models output instead (failure modes). Classifying every miss on an actual held-out value against the training value sets (pooled over the five instances) shows the failure mode is field- and scale-dependent. On held-out *allergies*, both anchors **omit** — they output “none” 100% of the time, a safe decline rather than a fabrication. On held-out *complaints*, the 10M model **substitutes a memorized training complaint in 86% of misses** (258/300), while the 3.15M model instead produces garbled or novel strings (98% of misses); held-out *medications* draw

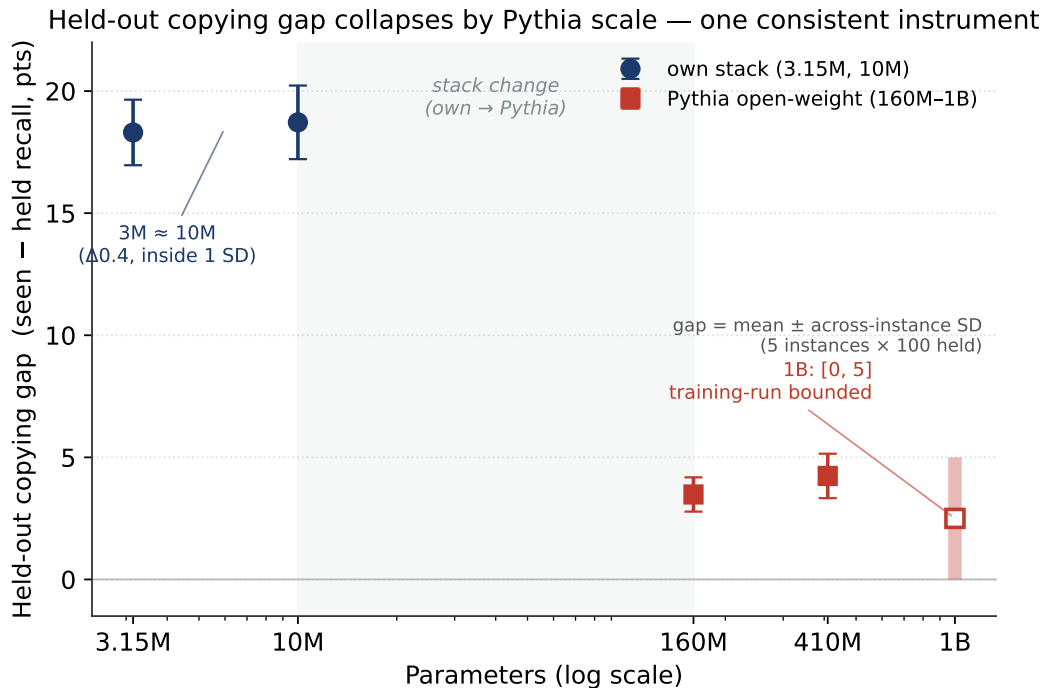


Figure 1: Held-out copying gap vs. scale on one consistent instrument. Own-stack anchors (3.15M, 10M) sit at ~ 18 pts; the tested Pythia rungs (160M, 410M) at 3.5–4.2 pts; 1B is a training-run–bounded interval $[0, 5]$. The shaded band marks the own→Pythia stack change — the x-axis is not a pure scale axis (§9).

omissions at 3M (66%) and garbled output at 10M. Substitution is the memorization signature in its most literal form — copying the wrong, *familiar* value in place of the right, *novel* one — and it **emerges with scale** (1% of nano misses vs. 54% of scale misses are substitutions), consistent with the 10M model having learned the complaint-slot distribution well enough for that prior to override the input. We report this as behavioral characterization; the circuit-level account is deferred to the mechanism stage (§10).

8.2 What the result does and does not establish

Within Pythia the gap is already small at 160M with no clean monotonic trend — and the fieldwise re-score explains why: the 160M→410M flatness is a *single-field* residual (allergy) that scale does not repair, so the ladder captured a low-gap regime rather than a transition. Reconciling with Stage S: the $3.2\times$ step (3M→10M) did not move the gap ($18.3 \pm 1.3 \rightarrow 18.7 \pm 1.5$ — a 0.4-point change inside one SD, now measured rather than inferred from two single points); the $16\times$ step to 160M, **or** the stack change, coincides with a much smaller gap — these are not separated here. The formal band is not PERSISTS (the top-rung interval reaches 0, far from the ≥ 10 that PERSISTS requires); it is consistent with a small residual or none.

8.3 Two distinct lessons

The paper carries two takeaways that are worth keeping separate, because they are supported by different evidence and would survive independently.

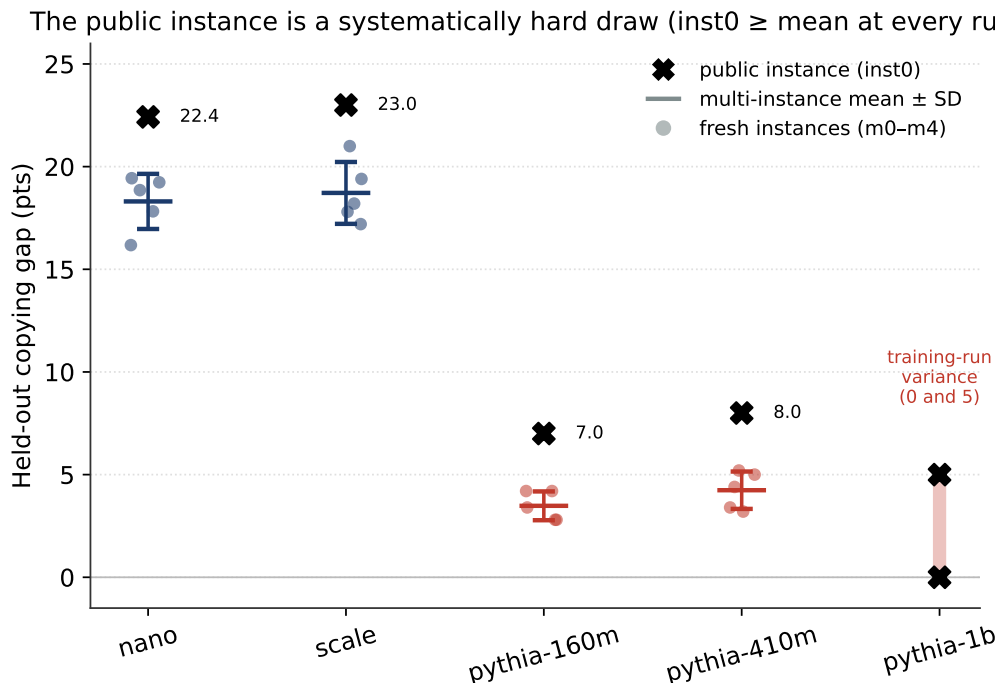


Figure 2: Per-rung instance difficulty. The single public instance (inst0 , $\times$) exceeds the five-instance mean $\pm$ SD at every rung; at 1B the two training runs (0 and 5) bracket the interval. This is why a single-instance anchor read higher than the powered mean.

What we learned about language models. The held-out copying gap is a *pronounced small-model phenomenon*: ~ 18 points in from-scratch models at 3–10M, essentially flat across that $3.2\times$ scale step, and only single-digit points in the tested Pythia models at 160M and above. The fieldwise breakdown makes the phenomenon precise rather than diffuse: the gap lives *entirely* in the open-vocabulary fields and is exactly zero in the closed-value fields, so what these small models fail at is specifically **copying a held-out lexical value into an open slot**, not being unfaithful in general — and the closed-value fields are a within-task control for that claim. Measured on the actual held-out values (§8.1, the value-level metric), the failure is not merely large but **near-total**: ~ 80 – 87 points aggregate, with $\approx 0\%$ recall on held-out allergy values (and on held-out medications at 3M) against $\sim 100\%$ on seen ones — the 18-point dialogue-level number substantially understates it. The misses are not random noise: at 10M the dominant error on held-out complaints is **substituting a memorized training complaint** (86% of misses), while held-out allergies are omitted rather than fabricated — memorization overriding the input in the most literal sense, and a failure mode that *changes shape* with scale (nano garbles where scale substitutes). The Pythia fieldwise re-score then sharpens the picture at the other end of the ladder: the residual Pythia gap is *entirely* the allergy slot — the one open slot with only five training values — where even 410M fails totally on the held-out value while having fully solved complaints and medications; on the slot-diversity hypothesis (§8.1), what the ladder measures is not one capability switching on but a per-slot competition between classify-and-memorize and copy, settled slot-by-slot by training diversity. What we do **not** get to say is *why* it shrinks with the ladder — the nano $\rightarrow$ Pythia step changes scale together with pretraining data, tokenizer, architecture, and finetuning method (§9), so the honest claim is “the gap largely disappears by the Pythia pipeline,” not “parameter count removes it.” The one clean causal statement is negative and comes from the controlled within-stack step: at this recipe, going

from 3M to 10M did not move the gap.

What we learned about measurement. Independently of the model result, estimating a gap like this reliably required two corrections that the data forced on us. (i) *Single-instance evaluation was underpowered, and worse, biased*: the public instance is not merely noisy but *systematically hard* — its gap exceeds the multi-instance mean at every rung — so a fixed public instance can skew an effect size even when it is content-addressed and reproducible. (ii) *The precision floor moves with the regime*: in the high-gap regime the dominant uncertainty is across evaluation instances, but by 1B the gap is small and the dominant uncertainty becomes the *training run* (two fixed-seed retrains split 5/0), so the right replication unit changes from eval-instance to training-seed. Both lessons are prescriptive for anyone measuring faithfulness gaps in small models and hold regardless of how the scale-vs-stack question is eventually resolved.

9 Limitations

- **Scale vs. stack confound (primary).** The nano→Pythia comparison changes parameter count *and* at least four other variables simultaneously, each a plausible alternative cause of the reduction: (i) **pretraining data quantity** ($\sim 200\text{M}$ own-stack tokens vs. Pythia’s $\sim 300\text{B}$ — a $\sim 1500\times$ difference; the own-stack is roughly compute-optimal for its size per Hoffmann et al., 2022, whereas Pythia is heavily over-trained), (ii) **tokenizer** (4098-vocab BPE vs. $\sim 50\text{k}$; larger vocabularies fragment field values like “ibuprofen” into fewer sub-tokens, which could itself change copy success), (iii) **architecture family**, and (iv) **finetuning method** (own-stack full fine-tune vs. Pythia LoRA $r=16$ — LoRA may preserve pretrained copying while suppressing the memorization pathway full FT exploits). We therefore claim only that the gap is much smaller under the Pythia *pipeline*, not that scale per se removes it. A pre-registered own-stack scale ladder is designed to separate scale from the stack bundle. (*Post-freeze addendum: that control has since been run once — an own-stack 160M model on the identical recipe and instrument retains a 16.9 ± 1.7 diluted gap, so the own-stack curve is flat across $50\times$ of scale where Pythia-160M reads 3.5, and the pre-registered decision rule fires **stack-dominant**. Two single-factor arms then each halve the gap identically — LoRA on the same checkpoint: 7.1 ± 1.2 ; Chinchilla-scaled pretraining (3.2B tokens) with full FT retained: 7.0 ± 1.0 — so the large gap is the interaction of an under-trained base with full-parameter adaptation, with data and method acting as substitutes. Single training run per cell; full report in follow-up work.*)
- **Transition point unobserved.** The gap drops somewhere between own-stack 10M (18.7) and Pythia-160M (3.5), but those endpoints are on different stacks; within Pythia the gap is already low at 160M with no monotonic trend, so “by Pythia scale” is safe while “as models scale up” is not yet supported.
- **1B point estimate.** Bounded, not identified, by training nondeterminism (§6.2).
- **Exact-match metric.** The gap is a difference of exact-match recalls; per Schaeffer et al. (2023) such metrics can exaggerate scale-linked transitions, which is part of why we report across-instance SD and, at 1B, a training-run interval.
- **Gap-definition subtleties (conservative direction).** The seen/held split is at the dialogue level, so the held bucket mixes in seen field-values (a “held” dialogue draws a held-out value only per-field-probabilistically); and recalls are computed over parsed dialogues only, where

at the nano rung the held split parses slightly less often than the seen split. Both push the *measured* gap toward zero, so reported gaps (aggregate and per-field) are conservative lower bounds on the pure held-out-value effect; neither biases the cross-model comparison, which uses the identical instrument.

- **Single task and single scale family.** One synthetic clinical-summarization task; one scaling family. Generality across domains and families is untested here.
- **Anchor precision (resolved).** The 3M/10M anchors were re-scored on the same multi-instance instrument as the rest of the ladder (18.3 ± 1.3 , 18.7 ± 1.5); the whole ladder now shares one instrument, and the anchor gaps carry across-instance SD like the Pythia rungs.

10 Future work (axes, not replacements)

The program extends as a matrix of one-variable-at-a-time questions, keeping Pythia as the reference scaling family: **Stage F** (family) holds capability $\sim$ fixed ($\sim$ 160–500M) and varies the family (Pythia vs. SmolLM/OLMo/TinyLlama) to ask whether the reduction tracks size or pipeline — the direct test of §9’s primary confound, alongside the pre-registered own-stack scale ladder. **Stage O** (objective) varies the finetuning objective at fixed model; **Stage R** (retrieval) adds RAG/pointer/copy-head mechanisms. **Frontier models** belong as *external validation* — “does the phenomenon still appear in production-grade systems?” — not as ladder rungs, since they change every variable at once. **Stage M** (mechanism) is deliberately deferred until the behavioral phenomenon is fully deconfounded, so it asks “why does this residual occur?” rather than carrying the burden of justifying the ladder; the natural hypotheses are copy/retrieval circuitry — induction heads (Olsson et al., 2022) and retrieval heads (Wu et al., 2024) — that generalize poorly from seen to held-out values, with the behavioral failure-mode split (omission vs. memorized substitution, §8.1) as its starting evidence.

11 Conclusion

A severe held-out copying gap in sub-10M models — specifically a failure to copy held-out lexical values into open slots, with the closed-value fields a zero-gap internal control — was substantially smaller in the tested Pythia models; the cause (scale vs. pipeline) is left open by design. The measurement story is inseparable from the result: single-instance evaluation was under-powered, and at the top of the ladder training nondeterminism — not evaluation noise — sets the precision floor. Both were surfaced by pre-registered checks (equivalence, determinism) that failed productively. The contribution is less any single number than a research unit that repeatedly incorporated its own measurement limitations into its conclusions rather than explaining them away.

Reproducibility. All eval instances, recipes, scorers, per-run result files, and library versions are content-addressed and archived in the public repository (<https://github.com/wwk5q8z6kk-bit/nano-lm>); every number in this paper regenerates from committed artifacts.

References

Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, et al. Pythia: A suite for analyzing large language models across training and scaling. In *International Conference on Machine Learning (ICML)*, 2023. arXiv:2304.01373.

- Nelson Elhage, Neel Nanda, Catherine Olsson, et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 2021.
- Jiatao Gu, Zhengdong Lu, Hang Li, and Victor O.K. Li. Incorporating copying mechanism in sequence-to-sequence learning. In *Proceedings of ACL*, 2016. arXiv:1603.06393.
- Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, et al. Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*, 2022.
- Ziwei Ji, Nayeon Lee, Rita Frieske, et al. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12), 2023. arXiv:2202.03629.
- Jared Kaplan, Sam McCandlish, Tom Henighan, et al. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.
- Joshua Maynez, Shashi Narayan, Bernd Bohnet, and Ryan McDonald. On faithfulness and factuality in abstractive summarization. In *Proceedings of ACL*, 2020.
- Catherine Olsson, Nelson Elhage, Neel Nanda, et al. In-context learning and induction heads. *Transformer Circuits Thread*, 2022. arXiv:2209.11895.
- Hung Viet Pham, Shangshu Qian, Jiannan Wang, Thibaud Lutellier, Jonathan Rosenthal, Lin Tan, Yaoliang Yu, and Nachiappan Nagappan. Problems and opportunities in training deep learning software systems: An analysis of variance. In *Proceedings of ASE*, 2020.
- David Picard. torch.manual_seed(3407) is all you need: On the influence of random seeds in deep learning architectures for computer vision. *arXiv preprint arXiv:2109.08203*, 2021.
- Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo. Are emergent abilities of large language models a mirage? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2304.15004.
- Abigail See, Peter J. Liu, and Christopher D. Manning. Get to the point: Summarization with pointer-generator networks. In *Proceedings of ACL*, 2017. arXiv:1704.04368.
- Jason Wei, Yi Tay, Rishi Bommasani, et al. Emergent abilities of large language models. *Transactions on Machine Learning Research (TMLR)*, 2022. arXiv:2206.07682.
- Wenhao Wu, Yizhong Wang, Guangxuan Xiao, Hao Peng, and Yao Fu. Retrieval head mechanistically explains long-context factuality. *arXiv preprint arXiv:2404.15574*, 2024.
- Cheng Xu, Shuhao Guan, Derek Greene, and M-Tahar Kechadi. Benchmark data contamination of large language models: A survey. *arXiv preprint arXiv:2406.04244*, 2024.
- Donglin Zhuang, Xingyao Zhang, Shuaiwen Leon Song, and Sara Hooker. Randomness in neural network training: Characterizing the impact of tooling. In *Proceedings of MLSys*, 2022. arXiv:2106.11872.